

2013 classification criteria for systemic sclerosis: an American college of rheumatology/European league against rheumatism collaborative initiative

Frank van den Hoogen,¹ Dinesh Khanna,² Jaap Fransen,³ Sindhu R Johnson,⁴ Murray Baron,⁵ Alan Tyndall,⁶ Marco Matucci-Cerinic,⁷ Raymond P Naden,⁸ Thomas A Medsger Jr,⁹ Patricia E Carreira,¹⁰ Gabriela Riemekasten,¹¹ Philip J Clements,¹² Christopher P Denton,¹³ Oliver Distler,¹⁴ Yannick Allanore,¹⁵ Daniel E Furst,¹² Armando Gabrielli,¹⁶ Maureen D Mayes,¹⁷ Jacob M van Laar,¹⁸ James R Seibold,¹⁹ Laszlo Czirjak,²⁰ Virginia D Steen,²¹ Murat Inanc,²² Otylia Kowal-Bielecka,²³ Ulf Müller-Ladner,²⁴ Gabriele Valentini,²⁵ Douglas J Veale,²⁶ Madelon C Vonk,³ Ulrich A Walker,⁶ Lorinda Chung,²⁷ David H Collier,^{28,29} Mary Ellen Csuka,³⁰ Barri J Fessler,³¹ Serena Guiducci,⁷ Ariane Herrick,^{32,33} Vivien M Hsu,³⁴ Sergio Jimenez,³⁵ Bashar Kahaleh,³⁶ Peter A Merkel,^{37,38} Stanislav Sierakowski,²³ Richard M Silver,³⁹ Robert W Simms,³⁸ John Varga,⁴⁰ Janet E Pope⁴¹

Handling editor Tore K Kvien

For numbered affiliations see end of article.

Correspondence to

Dr Frank van den Hoogen, Department of Rheumatology, St. Maartenskliniek, PO Box 9011, Nijmegen 6500 GM, The Netherlands; F.vandenHoogen@maartenskliniek.nl or Janet E Pope, St. Joseph's Health Care London, 268 Grosvenor Street, London, ON, Canada N6A 4V2; janet.pope@sjhc.london.on.ca.

This article is published simultaneously in the November 2013 issue of *Arthritis & Rheumatism*.

Accepted 9 August 2013

ABSTRACT

Objective The 1980 American College of Rheumatology (ACR) classification criteria for systemic sclerosis (SSc) lack sensitivity for early SSc and limited cutaneous SSc. The present work, by a joint committee of the ACR and the European League Against Rheumatism (EULAR), was undertaken for the purpose of developing new classification criteria for SSc.

Methods Using consensus methods, 23 candidate items were arranged in a multicriteria additive point system with a threshold to classify cases as SSc. The classification system was reduced by clustering items and simplifying weights. The system was tested by (1) determining specificity and sensitivity in SSc cases and controls with scleroderma-like disorders, and (2) validating against the combined view of a group of experts on a set of cases with or without SSc.

Results It was determined that skin thickening of the fingers extending proximal to the metacarpophalangeal joints is sufficient for the patient to be classified as having SSc; if that is not present, seven additive items apply, with varying weights for each: skin thickening of the fingers, fingertip lesions, telangiectasia, abnormal nailfold capillaries, interstitial lung disease or pulmonary arterial hypertension, Raynaud's phenomenon, and SSc-related autoantibodies. Sensitivity and specificity in the validation sample were, respectively, 0.91 and 0.92 for the new classification criteria and 0.75 and 0.72 for the 1980 ACR classification criteria. All selected cases were classified in accordance with consensus-based expert opinion. All cases classified as SSc according to the 1980 ACR criteria were classified as SSc with the new criteria, and several additional cases were now considered to be SSc.

Conclusions The ACR/EULAR classification criteria for SSc performed better than the 1980 ACR criteria for SSc and should allow for more patients to be classified correctly as having the disease.

INTRODUCTION

Systemic sclerosis (SSc; scleroderma) is a heterogeneous disease whose pathogenesis is characterised by three hallmarks: small vessel vasculopathy, production of autoantibodies, and fibroblast dysfunction leading to increased deposition of extracellular matrix.¹ The clinical manifestations and the prognosis of SSc vary, with the majority of patients having skin thickening and variable involvement of internal organs. Subsets of SSc can be discerned, that is, limited cutaneous SSc, diffuse cutaneous SSc, and SSc without skin involvement.¹

In the absence of a diagnostic test proving the absence or presence of SSc, several sets of classification criteria have been developed.^{2–6} The purpose of classification criteria is to enrol, in research studies, patients who in fact have the disease as determined using a uniform definition.⁷ Classification criteria are not synonymous with diagnostic criteria but will almost always mirror the list of criteria that are used for diagnosis.⁷ However, classification criteria generally are more standardised and less inclusive than physician diagnosis.

The current standard classification criteria for SSc are the 1980 preliminary criteria for the classification of systemic sclerosis (scleroderma), developed by the American College of Rheumatology (ACR).^{2–4} These classification criteria were developed using patients with longstanding SSc. As a consequence, patients with early SSc and ~20% of patients with limited cutaneous disease do not meet the criteria and are excluded from clinical studies.¹ Since the development of the 1980 criteria, knowledge regarding SSc-related autoantibodies has improved.^{11–13} In addition, characteristic nailfold capillary changes have been found to be associated with SSc, and nailfold capillaroscopy is widely accepted as a diagnostic tool.¹⁰ In 1988, LeRoy *et al*¹¹ proposed new criteria that

To cite: van den Hoogen F, Khanna D, Fransen J, *et al*. *Ann Rheum Dis* 2013;**72**:1747–1755.

included clinical features, autoantibodies, and capillaroscopy results, highlighting the differences between the two main SSc subsets. In 2001, LeRoy and Medsger proposed to revise the classification criteria to include 'early' SSc, using nailfold capillary pattern and SSc-related autoantibodies.⁶ It also has been demonstrated that the addition of nailfold capillary abnormalities and telangiectasias to the ACR SSc criteria improves their sensitivity.^{9 18}

Because of the insufficient sensitivity of the 1980 criteria and advances in knowledge about SSc, the ACR and the European League Against Rheumatism (EULAR) established a committee to provide a joint proposal for new classification criteria for SSc. The aims were to develop criteria that (1) encompass a broader spectrum of SSc including patients whose disease is in the early stage as well as those in the late stage; (2) include vascular, immunologic, and fibrotic manifestations; (3) are feasible to use in daily clinical practice; and (4) are in accordance with criteria used for diagnosis of SSc in clinical practice.⁷ These criteria are intended to be used by rheumatologists, researchers, national and international drug agencies, pharmaceutical companies, or any others involved in studies of SSc. Our objective was to develop a set of criteria that would enable identification of individuals with SSc for inclusion in clinical studies, being more sensitive and specific than previous criteria.

METHODS

Overview

The development and testing of the classification system for SSc was based on both data and expert clinical judgment. First, candidate items for the classification criteria were generated using consensus methods and evaluated using existing databases.^{19 20} Second, multicriteria decision analysis was used to reduce the number of candidate criteria and assign preliminary weights.²¹ The classification system was repeatedly tested and adapted using prospectively collected SSc cases and non-SSc controls, and compared against expert clinical judgment. Third, the classification criteria were tested in a validation cohort and tested against preexisting criteria sets.

Item generation and reduction

One hundred sixty-eight candidate criteria were identified through two Delphi exercises. A 3-round Delphi exercise and a face-to-face consensus meeting using nominal group technique facilitated reduction of the 168 items to 23.¹⁹ Using a random sample of existing databases (SSc (n=783) and control patients with diseases similar to SSc (n=1 071), all based on physician diagnosis), the candidate criteria were found to have good discriminative validity.²⁰

Item reduction and weighting

Draft classification system

A face-to-face meeting of four European and four North American SSc experts was held to further reduce items and assign preliminary weights using multicriteria decision analysis. The number of experts was limited in advance to 8, and they were invited based on geographic representation, knowledge from a scientific and a practical diagnostic viewpoint, and availability. At the meeting, the experts determined by consensus to which cases the criteria should be and should not be applied, and which items are sufficient to allow classification of a patient as having SSc (sufficient criteria). They then participated in a multicriteria decision analysis to further reduce the 23 items and assign preliminary weights.²¹ The experts were presented

hypothetical pairs of cases with 2 of the 23 items at a time (eg, Raynaud's phenomenon positive *and* abnormal nailfold capillaries absent vs Raynaud's phenomenon negative *and* abnormal nailfold capillaries present, all other manifestations being considered equal) and they were asked to individually vote electronically on which case of the pair was more likely to be SSc. The result of the votes was immediately presented. If there was no complete agreement among the experts, considerations were discussed and a second round of voting was conducted. As a result of the repeated choices between two alternative cases, items were ranked, and weights for the items were derived using 1000 Minds decision-making software.²¹ Additional details about the methods are available in ref. ²².

Initial threshold identification

The committee prepared summaries of 45 SSc cases, with a concentration of cases that were difficult to classify. These were presented to 22 SSc experts who classified the cases as definite SSc or not. The draft classification system derived from the multicriteria decision analysis was applied to the 45 cases, resulting in a score for each case. The ranking of cases by the SSc experts and the ranking of cases based on the scores provided with the draft scoring system were examined. Higher scores in the scoring system were expected to relate to a higher probability that the experts would classify the case as SSc. Using these results, an initial threshold score for SSc was identified.

Reduction and testing of iterative changes

In the next step, the committee reduced the number of items, simplified the weights, and modified the threshold score. First, data on the candidate items were prospectively collected at 13 SSc centres in North America and 10 in Europe, using standardised case record forms. Data from 368 consecutive patients with SSc (diagnosis based on physician opinion) were collected, of whom half were to have had SSc for a maximum of 2 years (based on the time from the first non-Raynaud's symptom) in order to include early SSc. Data from 237 consecutive control patients with a scleroderma-like disorder (eosinophilic fasciitis (also called Shulman's disease or diffuse fasciitis with eosinophilia), scleromyxedema, systemic lupus erythematosus, dermatomyositis, polymyositis, primary Raynaud's phenomenon, mixed connective tissue disease, undifferentiated connective tissue disease, generalised morphea, nephrogenic systemic sclerosis, and diabetic cheiroarthropathy) were also collected. From these 605 patients a random sample of 100 SSc cases and 100 controls (50% from North America and 50% from Europe) was selected to form the derivation sample. The remaining 268 cases and 137 controls formed the validation sample. Institutional research ethics board approval was obtained for the collection of patient data.

The committee then met and made iterative changes to the draft system, which they continually applied in real time to the derivation cohort derived as described above. Using the derivation cohort, the scoring system was simplified by removing items that occurred with low frequency or were redundant, by aggregating similar items and then transforming the weights to obtain single digits. The preliminary score threshold was adjusted to account for the weight simplification. The impact of all proposed changes was evaluated by assessing changes to sensitivity and specificity of the criteria in the derivation cohort. The reference standard to test the sensitivity and specificity was the diagnosis by the SSc expert who submitted the case(s) and control(s).

At the same time, the changes in the classification system were also tested in 38 difficult-to-classify cases. Consequently,

weights of some items were adjusted to align the scoring system with the reference standard formed by the opinions of the SSc experts as to which cases were to be classified as having SSc.

Validation

The final classification system was independently tested using the validation sample of SSc cases and controls. Sensitivity and specificity were calculated for the 1980 ACR preliminary classification criteria for SSc,³ the classification criteria proposed by LeRoy and Medsger in 2001,⁶ and the newly developed classification criteria. Exact binomial confidence limits were calculated for sensitivity and specificity. The ACR criteria and LeRoy/Medsger criteria were compared with the new criteria using 2×2 tables with McNemar's χ^2 test and continuity correction. The criteria sets were also tested separately using only the subgroup of patients with a disease duration of ≤ 3 years. Further, the classification system was validated against the expert consensus on the set of 38 selected cases.

RESULTS

Draft classification system

The experts concluded that 'skin thickening of the fingers of both hands extending proximal to the metacarpophalangeal joints' was sufficient to classify a subject as having SSc. Further, patients with 'skin thickening sparing the fingers' are classified as not having SSc. It was agreed that the criteria should be applied to any patient considered for inclusion in an SSc study, without further specifications. Items with relatively low weights were deleted, and items considered to be from a similar domain were clustered (eg, fingertip lesions encompasses ulcers and pitting scars; lung involvement encompasses interstitial lung disease and pulmonary hypertension). Using conjoint analysis, the number of items was reduced from 23 to 14 and all items were assigned weights. The 14 resulting items (with weights) were as follows: bilateral skin thickening of the fingers (sclerodactyly) (weighted 14 if distal to a proximal interphalangeal joint only, 22 if whole finger), puffy fingers (weighted 5), fingertip lesions (weighted 16 if pitting scars or 9 if digital ulcers), finger flexion contractures (weighted 16), telangiectasia (weighted 10), abnormal nailfold capillaries (weighted 10), calcinosis (weighted 12), Raynaud's phenomenon (weighted 13), tendon or bursal friction rubs (weighted 21), interstitial lung disease/pulmonary fibrosis (weighted 14), pulmonary arterial hypertension (weighted 12), scleroderma renal crisis (weighted 11), oesophageal dilation (weighted 7), and SSc-related autoantibodies (weighted 15 if anticentromere antibody present, anticentromere pattern seen on antinuclear antibody testing, or anti-topoisomerase I (also called anti-Scl-70) or anti-RNA polymerase III present).

Initial threshold identification

Comparison of the case ranking from the scoring system and by the experts revealed that most of the experts ($\geq 75\%$) considered the cases with a score of >55 (except for 1 case) to be SSc. Similarly, most experts ($\geq 88\%$) considered cases with a score of <40 not to be SSc. With scores between 40 and 55 there was more diversity of opinion. Thus, it was concluded that the initial threshold would be a score of ≥ 56 .

Reduction and testing of iterative changes

The 14 items in the scoring system were reduced to 9 while maintaining sensitivity and specificity in the derivation sample. The items deleted were finger flexion contractures, calcinosis, tendon or bursal friction rubs, renal crisis, and oesophageal

dilation. Puffy fingers or sclerodactyly were combined into one item, and pulmonary arterial hypertension and interstitial lung disease were also combined into one item, resulting in 7 items for the scoring system. In the derivation sample, with reduction of the 14 items to 7, the sensitivity and specificity were 0.93 and 0.94, respectively. Weights were simplified by dividing each weight by five and rounding to the nearest integer. The threshold for this simplified scoring system was determined to be 9. The resulting sensitivity and specificity were 0.97 and 0.88, respectively.

Weights were further adjusted to align the scoring system with the experts' opinions (SSc or not SSc) on each of the 38 difficult-to-classify cases. To improve the specificity of the classification criteria, an exclusionary criterion was added: patients with a diagnosis that better explains their manifestations should not be classified as having SSc. These revisions resulted in the correct classification of all patient profiles judged to have SSc by the majority of experts.

THE SSC CLASSIFICATION CRITERIA

The new classification criteria are presented in table 1. The table shows the one criterion that, if present, is sufficient for classification as SSc, the two exclusionary criteria, and the seven items with a combined threshold above which cases are classified as SSc. The classification criteria may be applied to patients who may have SSc and are being considered for inclusion in an SSc study. As noted above, they are *not to be applied* to patients who have a SSc-like disorder that better explains their manifestations; and patients with 'skin thickening sparing the fingers' are not classified as having SSc.

If a patient has skin thickening of the fingers of both hands that extends proximal to the metacarpophalangeal joints, the classification system assigns nine points for this one item alone, which is sufficient to classify the patient as having SSc with no further application of the point system needed. Otherwise, the point system is applied by adding the scores for manifestations that are 'positive.' The items are skin thickening of the fingers, fingertip lesions, telangiectasia, abnormal nailfold capillaries, pulmonary arterial hypertension and/or interstitial lung disease, Raynaud's phenomenon, and SSc-related autoantibodies. Two items, skin thickening of the fingers and fingertip lesions, include two different possible manifestations. If a patient has both manifestations, the score for the category is the higher score of the two manifestations. For example, in the item skin thickening of the fingers, if a patient has both manifestations, that is, puffy fingers (weighted 2) and sclerodactyly (weighted 4), the total score for the item would be 4. The maximum possible score is 19, and patients with a score of ≥ 9 are classified as having SSc. The definitions of the items used in the criteria are provided in table 2.

Validation

Table 3 shows the characteristics of the validation sample (268 patients with SSc, 137 controls). The sensitivity and specificity of the new SSc classification criteria were compared with those of the 1980 ACR classification criteria and the classification criteria proposed by LeRoy and Medsger, and the results are shown in table 4. The sensitivity and specificity of the new SSc criteria were, respectively, 0.91 and 0.92 in the validation sample. The new criteria performed better than the two earlier classification schemes in terms of sensitivity and specificity ($p=0.01$ vs the 1980 ACR criteria, $p=0.004$ vs the LeRoy/Medsger criteria). The area under the receiver operating characteristic curve of the classification system tested against presence

Table 1 The American College of Rheumatology/European League Against Rheumatism criteria for the classification of systemic sclerosis*

Item	Sub-item(s)	Weight/ score†
Skin thickening of the fingers of both hands extending proximal to the metacarpophalangeal joints (<i>sufficient criterion</i>)	–	9
Skin thickening of the fingers (<i>only count the higher score</i>)	Puffy fingers	2
	Sclerodactyly of the fingers (distal to the metacarpophalangeal joints but proximal to the proximal interphalangeal joints)	4
Fingertip lesions (<i>only count the higher score</i>)	Digital tip ulcers	2
	Fingertip pitting scars	3
Telangiectasia	–	2
Abnormal nailfold capillaries	–	2
Pulmonary arterial hypertension and/or interstitial lung disease (<i>maximum score is 2</i>)	Pulmonary arterial hypertension	2
	Interstitial lung disease	2
Raynaud's phenomenon	–	3
SSc-related autoantibodies (anticentromere, anti-topoisomerase I [anti-Scl-70], anti-RNA polymerase III) (<i>maximum score is 3</i>)	Anticentromere	3
	Anti-topoisomerase I	
	Anti-RNA polymerase III	

*These criteria are applicable to any patient considered for inclusion in a systemic sclerosis study. The criteria are not applicable to patients with skin thickening sparing the fingers or to patients who have a scleroderma-like disorder that better explains their manifestations (eg, nephrogenic sclerosing fibrosis, generalised morphea, eosinophilic fasciitis, sclerodema diabeticorum, scleromyxedema, erythromyalgia, porphyria, lichen sclerosis, graft-versus-host disease, diabetic cheiroarthropathy).

†The total score is determined by adding the maximum weight (score) in each category. Patients with a total score of ≥ 9 are classified as having definite systemic sclerosis. SSc, systemic sclerosis.

of SSc in the validation sample was 0.81 (95% CI 0.77 to 0.85). The performance of the new criteria in patients with disease of ≤ 3 years' duration was similar to the performance in the group overall (table 4).

The classification system was additionally tested against expert opinion (n=16 experts), using the set of 38 selected cases (table 5). All of the cases scoring ≥ 9 were considered SSc, whereas cases scoring < 9 were regarded as not being SSc or were controversial. With the proposed system all of these cases were classified in accordance with consensus-based expert opinion. All cases that were classified as SSc with the 1980 ACR criteria were also classified as SSc with the new criteria, as were several cases not classified as SSc with the 1980 ACR criteria.

DISCUSSION

A classification system for systemic sclerosis is needed to ensure that all patients categorised as having SSc for inclusion in studies do indeed have the disease, based on specific defined characteristics. The major reason to revise the 1980 ACR criteria was that those criteria lacked adequate sensitivity, especially in patients with early SSc or with limited cutaneous SSc.^{9 10 18} The proposed classification criteria are superior and exhibit greater sensitivity and specificity compared to the 1980 criteria and the classification criteria proposed by LeRoy and Medsger. All profiles of patients who were considered to have SSc by a majority of experts were indeed classified as SSc with the new classification system, and the new system is more inclusive and also performs well in patients with early disease (≤ 3 years since diagnosis).

Table 2 Definitions of items/sub-items in the American College of Rheumatology/European League Against Rheumatism criteria for the classification of systemic sclerosis

Item	Definition
Skin thickening	Skin thickening or hardening not due to scarring after injury, trauma, etc.
Puffy fingers	Swollen digits—a diffuse, usually nonpitting increase in soft tissue mass of the digits extending beyond the normal confines of the joint capsule. Normal digits are tapered distally with the tissues following the contours of the digital bone and joint structures. Swelling of the digits obliterates these contours. Not due to other causes such as inflammatory dactylitis.
Fingertip ulcers or pitting scars	Ulcers or scars distal to or at the proximal interphalangeal joint not thought to be due to trauma. Digital pitting scars are depressed areas at digital tips as a result of ischaemia, rather than trauma or exogenous causes.
Telangiectasia	Telangiectasiae are visible macular dilated superficial blood vessels, which collapse upon pressure and fill slowly when pressure is released. Telangiectasiae in a scleroderma-like pattern are round and well demarcated and found on hands, lips, inside of the mouth, and/or are large mat-like telangiectasiae. Distinguishable from rapidly filling spider angiomas with central arteriole and from dilated superficial vessels.
Abnormal nailfold capillary pattern consistent with systemic sclerosis	Enlarged capillaries and/or capillary loss with or without pericapillary haemorrhages at the nailfold. May also be seen on the cuticle.
Pulmonary arterial hypertension	Pulmonary arterial hypertension diagnosed by right-sided heart catheterisation according to standard definitions.
Interstitial lung disease	Pulmonary fibrosis seen on high-resolution CT or chest radiography, most pronounced in the basilar portions of the lungs, or occurrence of 'Velcro' crackles on auscultation, not due to another cause such as congestive heart failure.
Raynaud's phenomenon	Self-reported or reported by a physician, with at least a 2-phase colour change in finger(s) and often toe(s) consisting of pallor, cyanosis, and/or reactive hyperemia in response to cold exposure or emotion; usually one phase is pallor.
SSc-related auto antibodies	Anticentromere antibody or centromere pattern seen on antinuclear antibody testing, anti-topoisomerase I antibody (also known as anti-Scl-70 antibody), or anti-RNA polymerase III antibody. Positive according to local laboratory standards.

SSc, systemic sclerosis.

Table 3 Characteristics of the derivation sample and the validation sample*

Item	Derivation sample		p Value	Validation sample		p Value
	SSc (n=100)	Scleroderma-like disorder (n=100)		SSc (n=268)	Scleroderma-like disorder (n=137)	
Age, mean±SD years	55±13	51±15	0.05	54±13	52±15	0.17
Female	86 (86)	79 (79)	0.25	221 (83)	101 (75)	0.08
Region						
North America	50	50	—	191 (68)	91 (32)	0.32
Europe	50	50		77 (63)	46 (37)	
Time since onset of Raynaud's phenomenon, median (IQR) years	13 (7–18)	12 (4–18)	0.42	9 (5–18)	10 (4–22)	0.40
Time since first non-Raynaud's symptom, median (IQR) years	10 (4–13)	9 (2–14)	0.58	7 (3–12)	7 (3–15)	0.89
Time since diagnosis, median (IQR) years	8 (3–12)	6 (1–9)	0.10	5 (2–11)	4 (1–7)	0.016
Scleroderma-like disorders						
Systemic lupus erythematosus	—	28 (28)	—	—	32 (23)	—
Polymyositis/dermatomyositis	—	23 (23)	—	—	21 (15)	—
Primary Raynaud's phenomenon	—	19 (19)	—	—	7 (5)	—
Mixed connective tissue disease	—	9 (9)	—	—	14 (10)	—
Undifferentiated connective tissue disease	—	8 (8)	—	—	17 (12)	—
Eosinophilic fasciitis	—	6 (6)	—	—	16 (12)	—
Nephrogenic sclerosing fibrosis	—	3 (3)	—	—	3 (2)	—
Generalised morphea	—	5 (5)	—	—	8 (6)	—
Scleromyxedema	—	1 (1)	—	—	3 (2)	—
Graft-versus-host disease	—	3 (3)	—	—	3 (2)	—
Other diagnoses	—	8 (8)	—	—	13 (9)	—
Manifestations†						
Raynaud's phenomenon	91 (91)	49 (49)	<0.0001	257 (96)	63 (46)	<0.0001
Autoantibodies	68 (68)	7 (7)	<0.0001	137 (51)	15 (11)	<0.0001
Anticentromere antibody	33 (33)	5 (5)	<0.0001	41 (15)	8 (6)	0.0057
Anti-topoisomerase-I	34 (34)	1 (1)	<0.0001	69 (26)	7 (5)	<0.0001
Anti-RNA polymerase III	2 (2)	1 (1)	1.0	27 (10)	0	<0.0001
Puffy fingers	61 (61)	17 (17)	<0.0001	169 (63)	24 (18)	<0.0001
Abnormal nailfold capillaries	54 (54)	24 (24)	<0.0001	146 (54)	51 (37)	0.0010
Dilated vessels	37 (37)	28 (28)	0.08	124 (46)	34 (25)	0.0080
Avascular areas	21 (21)	11 (11)	0.08	86 (32)	9 (7)	<0.0001
Haemorrhages	12 (12)	9 (9)	0.64	63 (24)	8 (6)	<0.0001
Digital tip ulcers	53 (53)	8 (8)	<0.0001	108 (40)	12 (9)	<0.0001
Pitting scars	53 (53)	5 (5)	<0.0001	105 (39)	5 (4)	<0.0001
PAH or ILD	48 (48)	14 (14)	<0.0001	138 (52)	14 (10)	<0.0001
PAH	44 (44)	10 (10)	<0.0001	20 (7)	2 (1)	0.012
ILD	12 (12)	4 (4)	0.037	131 (49)	12 (9)	<0.0001
Telangiectasia	35 (35)	10 (10)	<0.0001	68 (25)	13 (9)	0.0002
Skin thickening of fingers to proximal of MCP joints	26 (26)	1 (1)	<0.0001	105 (39)	6 (4)	<0.0001
Skin thickening of fingers distal to MCP joints	5 (5)	38 (38)	<0.0001	178 (66)	24 (18)	<0.0001

*Data were prospectively collected on 605 consecutive patients with systemic sclerosis (SSc) or a scleroderma-like disorder (see Methods); half of the SSc patients at each site were to have early SSc. A random sample of 100 SSc cases and 100 controls, 50% from North America and 50% from Europe, was selected to form the derivation sample, and the remaining patients formed the validation sample. Except where indicated otherwise, values are the number (%).

†Listed in order of frequency of occurrence in the SSc derivation sample.

ILD, interstitial lung disease; MCP, metacarpophalangeal; PAH, pulmonary arterial hypertension.

The newly developed classification system includes disease manifestations of the three hallmarks of SSc: fibrosis of the skin and/or internal organs, production of certain autoantibodies, and vasculopathy. The four items comprising the 1980 ACR classification criteria (scleroderma proximal to the metacarpophalangeal joints, sclerodactyly, digital pitting scars (not pulp loss), and bilateral basilar pulmonary fibrosis)³ are also included, as are the items in the criteria proposed in 2001 by LeRoy and Medsger (Raynaud's phenomenon, autoantibodies, nailfold capillaroscopy abnormalities, skin fibrosis).⁶

The new criteria include one criterion that alone is sufficient for classification as SSc: skin thickening of the fingers extending proximal to the metacarpophalangeal joints, which is similar to the 1980 criteria. If the single sufficient criterion is not fulfilled, the point system is applied and patients with a score of ≥ 9 are classified as having SSc. All items in the classification criteria represent measurements that are performed in routine clinical practice. The criteria are meant for inclusion of SSc patients in studies, not for SSc diagnosis. Although the list of items in the classification criteria mimics the list of items one usually uses for diagnosis, in

Table 4 Sensitivity and specificity of the 2013 SSc classification criteria and previous SSc classification criteria, overall and in early SSc

	Derivation sample (n=200)		Validation sample (n=405)		Validation sample, disease duration ≤3 years (n=100)	
	Sensitivity (95% CI)	Specificity (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)	Sensitivity (95% CI)	Specificity (95% CI)
1980 ACR SSc criteria	0.80 (0.72 to 0.87)	0.77 (0.68 to 0.84)	0.75 (0.70 to 0.80)	0.72 (0.64 to 0.79)	0.75 (0.70 to 0.80)	0.72 (0.63 to 0.79)
2001 LeRoy/Medsger SSc criteria	0.76 (0.68 to 0.84)	0.69 (0.68 to 0.84)	0.75 (0.70 to 0.80)	0.78 (0.70 to 0.85)	0.80 (0.69 to 0.88)	0.76 (0.53 to 0.92)
2013 ACR/EULAR SSc criteria	0.95 (0.90 to 0.98)	0.93 (0.86 to 0.97)	0.91 (0.87 to 0.94)	0.92 (0.86 to 0.96)	0.91 (0.83 to 0.96)	0.90 (0.70 to 0.99)

ACR, American College of Rheumatology; EULAR, European League Against Rheumatism; SSc, systemic sclerosis.

practice the diagnosis of SSc may also be informed by items not included in the classification criteria, such as tendon friction rubs, calcinosis, and dysphagia. Consequently, patients *classified* as having SSc are a subset of patients being *diagnosed* as having SSc, with diagnosis being more sensitive. Ideally, there would be no difference between diagnosis and classification criteria.

As intended, the new classification system incorporates the considerable advances made in the diagnosis of SSc. It includes the concept of specific serum autoantibodies such as anti-topoisomerase I, anticentromere, and anti-RNA polymerase III.^{15–23} There is the possibility that testing for additional SSc autoantibodies, such as anti-Th/To, anti-U3 RNP, and others, may become more widely available. The criteria also acknowledge the value of magnified nailfold visualisation in the diagnosis of SSc.^{14–15} Although capillaroscopy can be performed with highly specialised equipment such as videocapillaroscopy cameras, simple in-office ophthalmoscopes or dermatoscopes suffice for distinguishing between normal and abnormal nailfold capillaries.^{24–25} Capillaroscopy is now widely used, and considering the value of magnified nailfold visualisation in the diagnosis and management of SSc, these new criteria may encourage acquisition of this skill by physicians caring for SSc patients. Likewise, criteria for pulmonary artery hypertension have changed over the years. The ACR/EULAR committee recognises this, and the diagnosis of pulmonary artery hypertension should be based on the most recent accepted criteria from right-sided heart catheterisation.

Several items that are useful for recognising SSc in clinical practice, such as calcinosis, flexion contractures of the fingers, tendon or bursal friction rubs, renal crisis, oesophageal dilatation, and dysphagia are not included in the criteria. These were considered but did not substantially improve sensitivity or specificity. For example, renal crisis is a strong indicator of SSc, but its low frequency of occurrence makes it less useful for the purpose of classification.²⁰ The committee considered a non-point-based additive system,⁸ such as the ACR systemic lupus erythematosus criteria²⁶ or the 1980 ACR SSc criteria. We concluded, however, that assigning weights yielded superior results for SSc classification. Indeed, the weights were simplified to single-digit numbers to make the system easy to use even in the absence of a computing device. Similar weighted systems have been used for other rheumatic diseases.²⁷ The committee also decided not to include 'probable' or 'possible' SSc in the classification.

Examples of profiles not captured by the 1980 ACR criteria that fulfilled the new classification criteria are combinations of skin thickening of the whole finger, SSc-related autoantibodies, and pulmonary arterial hypertension and/or Raynaud's phenomenon. A patient with Raynaud's phenomenon, autoantibodies, and abnormal nailfold capillaries is not classified as having SSc, although such a patient may develop SSc over subsequent years.^{6–15}

Patients may have disease manifestations similar to those of SSc that are better explained by another well-defined disorder,

such as nephrogenic sclerosing fibrosis, generalised morphea, eosinophilic fasciitis, scleredema diabeticorum, scleromyxedema, porphyria, lichen sclerosis, graft-versus-host disease, or diabetic cheiroarthropathy. We decided it was not necessary to develop criteria that differentiated SSc specifically from these conditions. Patients with some of these diseases were included in the validation cohort of patients with SSc-like disorders, and it is possible that specificity may have been slightly higher had they been excluded.

In developing the revised SSc classification criteria, we followed the recommendations and guidelines of the ACR and EULAR, which included (1) collaboration between clinical experts and clinical epidemiologists in criteria development, (2) evaluation of the psychometric properties of each candidate criterion, and (3) description of the test sample (origin of the patients and control subjects).^{7–28} Ideally, phases of criteria development should have a balance between expert opinion and data-driven methods; yet, there should be avoidance of circularity of reasoning (a bias that can occur when the same experts developing the criteria are the ones contributing cases and comparison patients).²⁹ We included different experts at different steps in the development of the SSc criteria, to avoid circularity.

Testing and validating a classification system for SSc is difficult because there is no gold standard for defining a particular case as SSc; that is, there is no incontrovertible test or criterion. We relied on expert opinion for our gold standard, which is similar to the process used in the development of other criteria.⁸

In the absence of a gold standard, we developed and tested the proposed classification system against two standards of expert opinion: (1) the opinion of the clinician who selected cases for the North American and European derivation and validation cohorts, and (2) the combined opinion of a group of clinical experts in SSc. Both standards have strengths and weaknesses. Each individual clinician who selected cases had access to information that could have included aspects not captured by the forms, which were restricted to 23 particular manifestations. Data were obtained from several sites in Europe and North America, so this should improve generalisability and reduce selection bias. However, it is possible that other expert clinicians may have had a different opinion about particular cases. The consensus opinion of a group of experts who had the opportunity to discuss controversial cases strengthens the combined expert opinion. However, the group may not have been aware of some relevant information not included in the available data. It is also difficult for a group of experts to consider hundreds of cases in depth; however, this was managed by having the group consider in depth only those cases, or combinations of items, that appeared to be controversial. In this way, the expert group was able to form a consensus over the whole range of cases in the database. A key strength of the present work is the use of both standards for testing and validation of the proposed system.

Table 5 Expert consensus versus classification systems for selected case profiles*

No.	Skin thickening of the fingers	Finger-tips	Telangiectasia	Abnormal nailfold capillaries	Puffy fingers	Raynaud's phenomenon	ILD/pulmonary fibrosis	PAH	SSc-related autoantibodies	Classified as SSc	Score	1980 ACR criteria fulfilled	No. of experts agreeing (n=16)
1	Whole finger	Pitting scars	–	Yes	Yes	Yes	–	Yes	Yes	Yes	17	Yes	16
2	Whole finger	Pitting scars	–	–	Yes	Yes	Yes	–	Yes	Yes	15	Yes	16
3	Whole finger	Pitting scars	–	–	Yes	Yes	–	–	Yes	Yes	13	Yes	16
4	Whole finger	Pitting scars	Yes	Yes	–	Yes	–	–	–	Yes	12	Yes	15
5	Whole finger	–	–	Yes	Yes	Yes	–	–	Yes	Yes	12	No	15
6	Whole finger	–	–	Yes	–	Yes	–	–	–	Yes	11	No	10
7	–	Pitting scars	–	–	Yes	Yes	Yes	–	–	Yes	10	?	NA
8	–	Ulcers	–	–	–	Yes	Yes	–	Yes	Yes	10	No	14
9	–	–	–	Yes	–	Yes	Yes	–	Yes	Yes	10	No	NA
10	–	–	–	Yes	Yes	Yes	–	–	Yes	Yes	10	No	NA
11	–	Ulcers	–	–	Yes	Yes	–	–	Yes	Yes	10	No	NA
12	–	–	–	Yes	Yes	Yes	–	–	Yes	Yes	10	No	NA
13	Whole finger	–	–	–	Yes	Yes	–	–	Yes	Yes	10	No	15
14	Whole finger	–	–	–	–	Yes	–	–	Yes	Yes	10	No	NA
15	Whole finger	–	–	Yes	–	Yes	–	–	–	Yes	9	No	NA
16	Whole finger	–	Yes	–	Yes	Yes	–	–	–	Yes	9	No	NA
17	Whole finger	Ulcers	–	–	Yes	Yes	–	–	–	Yes	9	No	NA
18	Whole finger	–	–	Yes	–	Yes	–	–	–	Yes	9	No	NA
19	Whole finger	–	–	Yes	Yes	Yes	–	–	–	Yes	9	No	9
20	Whole finger	–	–	Yes	–	Yes	–	–	–	Yes	9	No	NA
21	–	Pitting scars	–	–	–	Yes	Yes	–	–	No	8	?	NA
22	–	–	–	–	Yes	Yes	–	–	Yes	No	8	No	NA
23	–	Pitting scars	–	Yes	–	Yes	–	–	–	No	8	No	4
24	–	Pitting scars	Yes	–	–	Yes	–	–	–	No	8	No	2
25	–	–	–	Yes	–	Yes	–	–	Yes	No	8	No	NA
26	–	–	–	–	Yes	Yes	Yes	–	–	No	7	No	NA
27	–	–	–	Yes	Yes	Yes	–	–	–	No	7	No	NA
28	Distal to PIP joints	–	–	Yes	Yes	Yes	–	–	–	No	7	No	5
29	–	–	–	Yes	–	Yes	Yes	–	–	No	7	No	NA
30	Whole finger	–	–	–	–	Yes	–	–	–	No	7	No	NA
31	Whole finger	–	Yes	–	Yes	–	–	–	–	No	6	No	NA
32	Distal to PIP joints	–	–	–	–	Yes	–	–	Yes	No	6	No	8
33	–	–	–	–	–	Yes	–	–	Yes	No	6	No	0
34	–	–	–	–	–	–	Yes	–	Yes	No	5	No	NA

Continued

Table 5 Continued

No.	Skin thickening of the fingers	Finger-tips	Telangiectasia	Abnormal nailfold capillaries	Puffy fingers	Raynaud's phenomenon	ILD/pulmonary fibrosis	PAH	SSc-related autoantibodies	Classified as SSc	Score	1980 ACR criteria fulfilled	No. of experts agreeing (n=16)
35	-	-	-	-	-	Yes	-	Yes	-	No	5	No	0
36	-	-	-	Yes	-	Yes	-	-	-	No	5	No	1
37	-	-	-	Yes	-	-	-	-	Yes	No	5	No	0
38	-	-	-	Yes	-	-	-	-	-	No	2	No	0

*Profiles of 38 cases with likely systemic sclerosis (SSc) or an SSc-like disorder, with presence or absence of the manifestations in the newly developed American College of Rheumatology (ACR)/European League Against Rheumatism SSc classification criteria, are shown. The cases are listed in order of the points awarded using the classification system (Score column) and are classified as SSc if the score is ≥ 9 . For 18 of the cases, the number of experts agreeing that the case had SSc is shown. None of the cases had skin thickening of the fingers of both hands extending proximal to the metacarpophalangeal joints (the single criteria that, if present, immediately classifies the individual as having SSc). Whole finger is sclerodactyly of the entire finger but not proximal to the metacarpophalangeal joints. ILD, interstitial lung disease; PAH, pulmonary arterial hypertension; ? , inconclusive; NA, not applicable (experts were not asked); PIP, proximal interphalangeal.

Author: In the last column head you say the number of experts was 16, but I don't see this anywhere in the text. Please correct, clarify, or add to the text.

Comp.: Please insert an N-dash in the PAH column for patients 6 and 7. Thanks.

The approach we used has other strengths and limitations as well. The methodology was state of the art, with validation by data and by expert opinion at every step. Various methods used in the development process have already been described.^{19 20} The criteria have face validity, because the items are routinely assessed in daily clinical practice and also were included in other important SSc classification criteria sets. The criteria allow for new developments in for example, autoantibody testing availability and/or new identification of scleroderma-associated autoantibodies, or assessment of nailfold capillaries. Formal joint analysis to derive the weights associated with items improved the sensitivity and specificity of the items, as was found also in the development of the recent ACR/EULAR criteria for the classification of rheumatoid arthritis.³⁰

The criteria have not been validated in ethnic groups that are not common in North America and Europe. This will require further studies. Regarding clinical use, the number of items and weights may not be easy to remember, but wide availability and (electronic) applications can be developed. The SSc classification criteria steering committee and the expert consultants agreed that the criteria could allow for classification of patients with another rheumatic disease as also having SSc (eg, having both systemic lupus erythematosus and SSc, or rheumatoid arthritis and SSc, etc.). Although this is a possible limitation, it permits individual researchers to decide whether to include subjects who fulfill criteria for more than one rheumatic disease in any particular study.

CONCLUSIONS

The ACR/EULAR classification criteria for SSc perform better than 1980 ACR preliminary criteria in terms of both sensitivity and specificity. They are relatively simple to apply to individual subjects. These criteria may be endorsed as inclusion criteria for SSc studies. Validation in other populations is encouraged.

Author affiliations

¹St. Maartenskliniek and Radboud University Nijmegen Medical Centre, Nijmegen, The Netherlands

²University of Michigan, Ann Arbor, Michigan, USA

³Radboud University Nijmegen Medical Centre, Nijmegen, The Netherlands

⁴Toronto Western Hospital, Mount Sinai Hospital, and University of Toronto, Toronto, Ontario, Canada

⁵Jewish General Hospital and McGill University, Montreal, Quebec, Canada

⁶Felix Platter Spital and University of Basel, Basel, Switzerland

⁷University of Florence, Florence, Italy

⁸Auckland City Hospital and New Zealand Health Ministry, Auckland, New Zealand

⁹University of Pittsburgh School of Medicine, Pittsburgh, Pennsylvania, USA

¹⁰Hospital Universitario 12 de Octubre, Madrid, Spain

¹¹Charité University Medicine Berlin, German Rheumatology Research Center, and Leibniz Institute, Berlin, Germany

¹²University of California, Los Angeles, David Geffen School of Medicine, Los Angeles, USA

¹³Royal Free Hospital and University College London Medical School, London, UK

¹⁴University Hospital Zurich, Zurich, Switzerland

¹⁵Université Paris Descartes and Hôpital Cochin, AP-HP, Paris, France

¹⁶Clinica Medica Università Politecnica delle Marche, Ancona, Italy

¹⁷University of Texas Health Science Center, Houston, Texas, USA

¹⁸Newcastle University Medical School, Newcastle upon Tyne, UK

¹⁹Scleroderma Research Consultants, Avon, Connecticut, USA

²⁰University of Pécs Medical Center, Pécs, Hungary

²¹Georgetown University School of Medicine, Washington, DC, USA

²²Istanbul University, Istanbul, Turkey

²³Medical University of Białystok, Białystok, Poland

²⁴Justus-Liebig University Giessen and Kerckhoff Clinic, Bad Nauheim, Germany

²⁵Second University of Naples, Naples, Italy

²⁶St. Vincent's University Hospital, Dublin, Ireland

²⁷Stanford University School of Medicine, Stanford, California, USA

²⁸University of Colorado Medical School and Denver Health Medical Center, Denver, Colorado, USA

²⁹Amgen, Thousand Oaks, California, USA

- ³⁰ Medical College of Wisconsin, Milwaukee, Michigan, USA
³¹ University of Alabama at Birmingham, Birmingham, UK
³² University of Manchester, Manchester, UK
³³ Salford Royal National Health Service Foundation Trust, Salford, UK
³⁴ Rutgers Robert Wood Johnson Medical School, New Brunswick, New Jersey, USA
³⁵ Thomas Jefferson University, Philadelphia, Pennsylvania, USA
³⁶ University of Toledo, Toledo, Ohio, USA
³⁷ University of Pennsylvania, Philadelphia, Pennsylvania, USA
³⁸ Boston University School of Medicine, Boston, Massachusetts, USA
³⁹ Medical University of South Carolina, Charleston, South Carolina, USA
⁴⁰ Northwestern University Feinberg School of Medicine, Chicago, Illinois, USA
⁴¹ St. Joseph's Health Care London and University of Western Ontario, London, Ontario, Canada

This criteria set has been approved by the American College of Rheumatology (ACR) Board of Directors and the European League Against Rheumatism (EULAR) Executive Committee. This signifies that the criteria set has been quantitatively validated using patient data, and it has undergone validation based on an external data set. All ACR/ EULAR-approved criteria sets are expected to undergo intermittent updates. The ACR is an independent, professional, medical and scientific society which does not guarantee, warrant, or endorse any commercial product or service.

Contributors All authors were involved in drafting the article or revising it critically for important intellectual content, and all authors approved the final version to be published. Drs. FvdH and JEP had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Study conception and design: FvdH, DK, JF, SRJ, MB, AT, MM-C, RPN, GR, CPD, DEF, AG, MDM, JRS, LC, VDS, MI, UM-L, DJV, UAW, LC, AH, SS, RMS, JEP. Acquisition of data: FvdH, DK, SRJ, MB, AT, MM-C, TAM, PEC, GR, PJC, CPD, OD, YA, DEF, MDM, JRS, LC, VDS, MI, OK-B, GV, DJV, MCV, UAW, DHC, MEC, BJF, SG, VMH, SJ, BK, PAM, SS, RMS, RWS, JV, JEP. Analysis and interpretation of data: FvdH, DK, JF, SRJ, MB, AT, MM-C, RPN, TAM, GR, PJC, CPD, DEF, MDM, JMWL, VDS, MI, OK-B, GV, DJV, PAM, SS, JEP.

Funding Supported by the American College of Rheumatology and the European League Against Rheumatism. Dr Khanna's work was supported by the Scleroderma Foundation (New Investigator award) and the NIH (National Institute of Arthritis and Musculoskeletal and Skin Diseases grant K24-AR-063120). Dr Johnson's work was supported by the Canadian Institutes of Health Research (Clinician Scientist award) and the Norton-Evans Fund for Scleroderma Research.

Competing interests DK has received consulting fees from Bayer, Bristol-Myers Squibb, Digna Biotech, InterMune, Gilead, and Merck (less than \$10 000 each) and consulting and speaking fees from Actelion and United Therapeutics (more than \$10 000 each). MM-C has received consulting fees, speaking fees, and/or honoraria from Bristol-Myers Squibb, Pfizer, and Actelion (less than \$10 000 each). OD has received consulting fees, speaking fees, and/or honoraria from Bayer, Pfizer, Ergonex, Bristol-Myers Squibb, Sanofi-Aventis, Sinoxa Pharma, United BioSource, Medac, Swedish Orphan Biovitrum, Boehringer Ingelheim Pharma, Novartis, and Active Biotech (less than \$10 000 each) and from Actelion and 4D Science (more than \$10 000 each); he also has a patent for the use of microRNA-29 for the treatment of systemic sclerosis. YA has received consulting fees, speaking fees, and/or honoraria from Roche, Actelion, and Pfizer (less than \$10 000 each). DEF has received consulting fees, speaking fees, and/or honoraria from Actelion, Gilead, and Genentech/Roche (less than \$10 000 each). AG has received research grants from Actelion, Roche, Abbott, and Pfizer. MDM has received consulting fees, speaking fees, and/or honoraria from Actelion, Practice Point Communications, and Medteligence (less than \$10 000 each) and receives royalties from Oxford University Press for *The Scleroderma Book* and from Henry Stewart Talks for a video of a prerecorded lecture on scleroderma. JMWL has received honoraria from Roche, Genentech, Pfizer, Abbott, Menarini, and Miltenyi Biotech (less than \$10 000 each) and research grants from Roche/Genentech and Trubion Pharmaceuticals. MI has received consulting fees, speaking fees, and/or honoraria from Actelion Turkey, Bayer Turkey, and GlaxoSmithKline Turkey (less than \$10 000 each). OK-B has received consulting fees, speaking fees, and/or honoraria from Actelion and Pfizer (less than \$10 000 each). DJV has received consulting fees, speaking fees, and/or honoraria from Pfizer, UCB, MSD, and Roche (less than \$10 000 each). MCV has received consulting fees, speaking fees, and/or honoraria from Actelion, Pfizer, GlaxoSmithKline, and United Therapeutics (less than \$10 000 each). DHC owns stock or stock options in Amgen. AH has received consulting fees, speaking fees, and/or honoraria from Actelion (less than \$10 000 each). SS has received consulting fees, speaking fees, and/or honoraria from Roche, Pfizer, Abbott, MSD, GlaxoSmithKline, and Egis Pharmaceuticals (less than \$10 000 each). RMS has received consulting fees, speaking fees, and/or honoraria from Genentech and Actelion (less than \$10 000 each).

Provenance and peer review Commissioned; internally peer reviewed.

REFERENCES

- Wollheim FA. Classification of systemic sclerosis: visions and reality. *Rheumatology (Oxford)* 2005;44:1212–16.
- Masi AT, Medsger TA Jr, Rodnan GP, *et al.* Methods and preliminary results of the Scleroderma Criteria Cooperative Study of the American Rheumatism Association. *Clin Rheum Dis* 1979;5:27–79.
- Subcommittee for Scleroderma Criteria of the American Rheumatism Association Diagnostic and Therapeutic Criteria Committee. Preliminary criteria for the classification of systemic sclerosis (scleroderma). *Arthritis Rheum* 1980;23:581–90.
- Preliminary criteria for the classification of systemic sclerosis (scleroderma). *Bull Rheum Dis* 1981;31:1–6.
- Nadashkevich O, Davis P, Fritzler MJ. A proposal of criteria for the classification of systemic sclerosis. *Med Sci Monit* 2004;10:CR615–21.
- LeRoy EC, Medsger TA Jr. Criteria for the classification of early systemic sclerosis. *J Rheumatol* 2001;28:1573–6.
- Classification and Response Criteria Subcommittee of the American College of Rheumatology Committee on Quality Measures. Development of classification and response criteria for rheumatic diseases [editorial]. *Arthritis Rheum* 2006;55:348–52.
- Johnson SR, Goek ON, Singh-Grewal D, *et al.* Classification criteria in rheumatic diseases: a review of methodologic properties. *Arthritis Rheum* 2007;57:1119–33.
- Lonzetti LS, Joyal F, Raynaud JP, *et al.* Updating the American College of Rheumatology preliminary classification criteria for systemic sclerosis: addition of severe nailfold capillaroscopy abnormalities markedly increases the sensitivity for limited scleroderma [letter]. *Arthritis Rheum* 2001;44:735–6.
- Hachulla E, Launay D. Diagnosis and classification of systemic sclerosis. *Clin Rev Allergy Immunol* 2011;40:78–83.
- LeRoy EC, Black C, Fleischmajer R Jr, *et al.* Scleroderma (systemic sclerosis): classification, subsets and pathogenesis. *J Rheumatol* 1988;15:202–5.
- Avouac J, Fransen J, Walker UA, *et al.*; EUSTAR Group. Preliminary criteria for the very early diagnosis of systemic sclerosis: results of a Delphi consensus study from EULAR Scleroderma Trials and Research Group. *Ann Rheum Dis* 2011;70:476–81.
- Walker JG, Pope J, Baron M, *et al.* The development of systemic sclerosis classification criteria. *Clin Rheumatol* 2007;26:1401–9.
- Harper FE, Maricq HR, Turner RE, *et al.* A prospective study of Raynaud phenomenon and early connective tissue disease: a five-year report. *Am J Med* 1982;72:883–8.
- Koenig M, Joyal F, Fritzler MJ, *et al.* Autoantibodies and microvascular damage are independent predictive factors for the progression of Raynaud's phenomenon to systemic sclerosis: a twenty-year prospective study of 586 patients, with validation of proposed criteria for early systemic sclerosis. *Arthritis Rheum* 2008;58:3902–12.
- Maricq HR, Weinberger AB, LeRoy EC. Early detection of scleroderma-spectrum disorders by in vivo capillary microscopy: a prospective study of patients with Raynaud's phenomenon. *J Rheumatol* 1982;9:289–91.
- Cutolo M, Matucci-Cerinic M. Nailfold capillaroscopy and classification criteria for systemic sclerosis. *Clin Exp Rheumatol* 2007;25:663–5.
- Hudson M, Taillefer S, Steele R, *et al.* Improving the sensitivity of the American College of Rheumatology classification criteria for systemic sclerosis. *Clin Exp Rheumatol* 2007;25:754–7.
- Fransen J, Johnson SR, van den Hoogen F, *et al.* Items for developing revised classification criteria in systemic sclerosis: results of a consensus exercise. *Arthritis Care Res (Hoboken)* 2012;64:351–7.
- Johnson SR, Fransen J, Khanna D, *et al.* Validation of potential classification criteria for systemic sclerosis. *Arthritis Care Res (Hoboken)* 2012;64:358–67.
- Hansen P, Ombler F. A new method for scoring multi-attribute value models using pairwise rankings of alternatives. *J Multi-Crit Decis Anal* 2009;15:87–107.
- Johnson SR, Naden RP, Fransen J, *et al.* Systemic sclerosis classification criteria: developing methods for multi-criteria decision analysis [abstract]. *Arthritis Rheum* 2012;64(Suppl):S372.
- Steen VD. Autoantibodies in systemic sclerosis. *Semin Arthritis Rheum* 2005;35:35–42.
- Baron M, Bell M, Bookman A, *et al.* Office capillaroscopy in systemic sclerosis. *Clin Rheumatol* 2007;26:1268–74.
- Hudson M, Masetto A, Steele R, *et al.* for the Canadian Scleroderma Research Group. Reliability of widefield capillary microscopy to measure nailfold capillary density in systemic sclerosis. *Clin Exp Rheumatol* 2010;28:S36–41.
- Tan EM, Cohen AS, Fries JF, *et al.* The 1982 revised criteria for the classification of systemic lupus erythematosus. *Arthritis Rheum* 1982;25:1271–7.
- Aletaha D, Neogi T, Silman AJ, *et al.* 2010 rheumatoid arthritis classification criteria: an American College of Rheumatology/European League Against Rheumatism collaborative initiative. *Arthritis Rheum* 2010;62:2569–81.
- Dougados M, Gossec L. Classification criteria for rheumatic diseases: why and how? [editorial]. *Arthritis Rheum* 2007;57:1112–15.
- Felson DT, Anderson JJ. Methodological and statistical approaches to criteria development in rheumatic diseases. *Baillieres Clin Rheumatol* 1995;9:253–66.
- Neogi T, Aletaha D, Silman AJ, *et al.* The 2010 American College of Rheumatology/ European League Against Rheumatism classification criteria for rheumatoid arthritis: Phase 2 methodological report. *Arthritis Rheum* 2010;62:2582–91.